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This document outlines and breaks down the scoring algorithm used by Salud.

1. Overview

Algorithm Name: Salud Health Score

Purpose:

This algorithm is used to assess the healthiness of an individual food item by turning a food's Nutrition Facts label and ingredient list into a score from 0 (least healthy) to 100 (most healthy). The score is built in two layers:

- 1. Base Quality Score:** the same number for every person. It only asks: "is this food generally nutritious?"
- 2. Goal Adjustment:** a small nudge, capped in size, based on the individual's stated goal (for example, losing weight, gaining weight, or general health).

Because the adjustment is capped, a person's goal can move a food's score by a little, but it can never turn an unhealthy food into a "healthy" one, or the reverse. A cookie stays low for everyone; a person's goal only decides how low.

Notation used in this document:

Every formula below uses italic letters to stand in for a number, the same way a recipe uses "2 cups flour." A small table under each formula explains exactly what each letter means and where to find it (typically right on the Nutrition Facts label).

2. High-Level Description

Every score follows the same five steps. Nothing about a person's goal is allowed to enter until the last step.

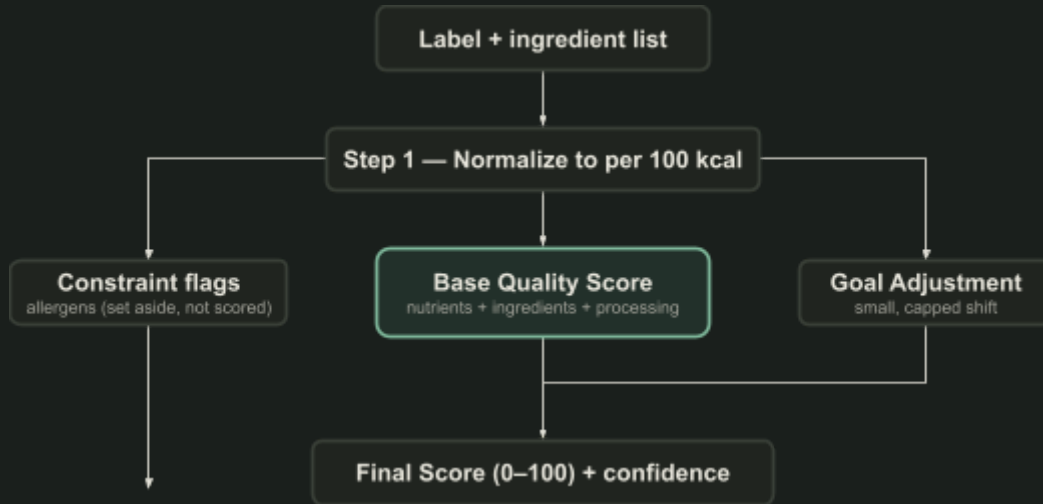


Fig. 1 – Allergens and dietary flags never touch the number; the goal only enters at the very last step, and only by a capped amount.

3. Algorithm Steps

This algorithm is broken into 8 parts

1. Normalize the label to “per 100 calories”
2. Constraint flags (set aside, not scored)
3. Nutrient Balance Score (N)
4. Food Ingredient Score (F)
5. Processing Score (S)
6. Combine into the Base Quality Score (B)
7. Goal Adjustment (Δ)
8. Final Score, band, and confidence

3.1 Normalize the label to “per 100 calories”

Nutrition labels list amounts per serving, but servings are different sizes for every food, which makes two foods hard to compare fairly. So the first step converts everything to per 100 calories instead of per serving. This also matches how nutrition scientists compare “the quality of the calories,” rather than penalizing a food just for being naturally more caloric per bite (like nuts or oil).

Type	Name	Where to find it	Symbol
Input	Calories per serving	Nutrition Facts label, “Calories”	C
Input	Any nutrient amount per serving	Nutrition Facts label (g or mg)	x
Output	Same nutrient, rescaled per 100 calories	Calculated	x_{100}

Formula 1 – normalize any nutrient

$$x_{100} = x \times (100 \div C)$$

Do this once for each of: fiber, added sugars, saturated fat, sodium, and protein. Keep calories per serving (C) handy since it is needed again in Step 6.

3.2 Constraint flags (set aside, not scored)

Before any math, check the ingredient list and label for hard facts that no score should average away. These become flags shown to the person (ex. “contains peanuts”) rather than points added or subtracted.

Flag	Where to find it
Allergens (milk, egg, peanut, tree nut, soy, wheat, fish, shellfish, sesame)	Label’s “Contains” statement
Diet compatibility (vegan, vegetarian, halal, kosher, gluten-free)	Ingredient list (look for animal products, gelatin, alcohol, or gluten-containing grains)

3.3 Nutrient Balance Score (N)

This piece rewards fiber and protein, and takes points away for added sugar, saturated fat, and sodium—the four nutrients most consistently linked to diet-related disease. Each of the five gets its own small score from 0 to 10 using the per-100-calorie amounts from Step 1.

Nutrient	Points formula (0–10)	Meaning
Fiber	$10 \times \min(x_{100} \div 9.5, 1)$	Full 10 points once fiber reaches 9.5 g per 100 kcal

Added sugar	$10 \times \max(1 - x_{100} \div 10, 0)$	Full 10 points at zero; 0 points at 10 g or more per 100 kcal
Saturated fat	$10 \times \max(1 - x_{100} \div 5, 0)$	Full 10 points at zero; 0 points at 5 g or more per 100 kcal
Sodium	$10 \times \max(1 - x_{100} \div 225, 0)$	Full 10 points at zero; 0 points at 225 mg or more per 100 kcal
Protein	$10 \times \min(x_{100} \div 8, 1)$	Full 10 points once protein reaches 8 g per 100 kcal

Formula 2 – Nutrient Balance Score

$$N = (P_{fiber} + P_{sugar} + P_{satfat} + P_{sodium} + P_{protein}) \div 5 \times 10$$

Add the five point values, divide by 5, and multiply by 10. The result, **N**, lands between 0 and 100.

3.4 Food Ingredient Score (**F**)

This piece looks at the first three items on the ingredient list (the ingredients present in the largest portion of the item since ingredients are listed by weight). Each is classified as whole (+1), refined or processed (−1), or neutral (0), using the reference list in the Appendix.

Position	Weight (w)	Why
1st ingredient	3	Largest amount in the food
2nd ingredient	2	Second largest
3rd ingredient	1	Third largest

Formula 3 – ingredient raw score

$R = \Sigma (w \times c)$ for the first three ingredients present

W = sum of the weights actually used (6 if there are 3+ ingredients, 5 if there are 2, 3 if there is 1)

Formula 4 – Food Ingredient Score

$$F = 50 + (R \div W) \times 50$$

50 is neutral. All-whole ingredients push it to 100; all-refined ingredients push it to 0.

3.5 Processing Score (S)

This piece looks at how far a food is from its whole-food state, using the widely used NOVA classification, then subtracts for specific flagged additives.

NOVA group	Description	Base points
1	Unprocessed or minimally processed (fruit, plain meat, milk, eggs)	100
2	Processed culinary ingredients (oil, butter, sugar, salt, sold on their own)	75
3	Processed foods (canned vegetables, cheese, fresh bread)	50
4	Ultra-processed (most packaged snacks, sodas, packaged cookies)	10

Then subtract 10 points for each of the following found in the ingredient list, up to a maximum deduction of 30:

Flag	Look for on the label
Artificial sweetener, color, or flavor	“Artificial flavor,” “Red 40,” “Sucralose,” etc.
Partially hydrogenated oil	“Partially hydrogenated [oil type]”
High-fructose corn syrup	“High fructose corn syrup”
MSG (added, not naturally occurring)	“Monosodium glutamate”

Formula 5 – Processing Score

$S = \max(\text{base points} - 10 \times n, 0)$ where n = number of flags found (capped so the deduction never exceeds 30)

3.6 Combine into the Base Quality Score (B)

The three pieces are combined with fixed weights: nutrients matter most, ingredients and processing share the rest evenly.



Fig. 2 – Why: nutrient content is the strongest predictor of health outcomes; ingredients and processing correct for cases nutrients alone miss (like a “low sugar” food full of additives).

Formula 6 – Base Quality Score

$$B = 0.7 \times N + 0.3 \times ((F + S) \div 2)$$

B is the score everyone sees before any personalization—the same number for every user, regardless of goal.

3.7 Goal Adjustment (Δ)

This is the only step where a person's goal enters. It looks at three properties of the food, and a table of “how much this goal cares” for each goal.

3.7.1 The three food properties

Property	Formula	Meaning
Energy fit (a_e)	$\text{clip}((D - 250) \div 250, -1, 1)$	D = calories per 100 g of food. Positive = calorie-dense; negative = calorie-light.
Protein fit (a_p)	$\text{clip}((x_{100,protein} - 4) \div 4, -1, 1)$	Positive = protein-rich relative to its calories
Sugar concern (a_s)	$\text{clip}((x_{100,protein} - 5) \div 5, -1, 1)$	Positive = notably sugary

“ $\text{clip}(\text{value}, -1, 1)$ ” means: if the value is above 1, use 1; if it’s below -1 , use -1 ; otherwise use it as is. It just keeps each property between -1 and 1.

D (calories per 100 g) comes from the label's serving size in grams: $D = C \div (\text{serving size in grams}) \times 100$.

3.7.2 How much each goal cares (λ values)

Goal	λ_e (energy)	λ_p (protein)	λ_s (sugar)
General health	-0.2	0.2	-0.3
Lose weight	-0.6	0.2	-0.3
Gain weight / build muscle	$+0.5$	0.5	-0.2

Notice the sign flips on energy: losing weight treats a calorie-dense food as a mark against it, while gaining weight treats the same density as a small point in its favor. Nothing else about the food changes only how much this one property is allowed to matter.

3.7.3 Combining into the adjustment

Formula 7 – Goal Adjustment

$$\Delta = 12 \times \tanh(\lambda_e a_e + \lambda_p a_p + \lambda_s a_s)$$

3.8 Final Score, band, and confidence

Formula 8 – Final Score

$$Score = clip(B + \Delta, 0, 100)$$

Band	Score range
Poor	0-39
Fair	40-59
Good	60-79
Excellent	80-100

If any nutrient on the label is missing or shows “o g” ambiguously, note it (the score is still computed but should be shown as an estimate rather than exact).